State-space Models, Transition Matrices, Controllability and Observability

Optimal Control, Homework 1

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The problems come from Chapter 1 of Kirk, Optimal Control Theory.

Problem 1-1 (series RLC circuit). From KVL and KCL, with $x_1 = i_L$ and $x_2 = v_C$, I wrote $\dot{x} = Ax + Be$ with $A = \begin{bmatrix} -R/L & -1/L \\ 1/C & 0 \end{bmatrix}$ and $B = [1/L;\ 0]$. For $L = 1$, $C = \frac{1}{2}$, $R = 3$ I computed $\Phi(s) = (sI - A)^{-1}$ and $\Phi(t) = \mathcal{L}^{-1}\{\Phi(s)\}$, a combination of e^{-t} and e^{-2t}. I then found the response to a rectangular input pulse on $1 \le t \le 2$ with zero initial conditions, writing $i_L(t)$ and $v_C(t)$ with shifted step functions.

Problem 1-11 (properties of the transition matrix). For time-varying systems I proved $\Phi(t, t) = I$, $\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0)$, $\Phi^{-1}(t_2, t_1) = \Phi(t_1, t_2)$ and $\frac{d}{dt}\Phi(t, t_0) = A(t)\Phi(t, t_0)$. I then specialised them to the time-invariant case $\Phi(t - t_0) = e^{A(t-t_0)}$.

Problem 1-13 (realisations from transfer functions). For $5/(s(s+1))$, $1/s^2$ and $5(s+2)/(s(s+1))$ I drew signal-flow graphs, wrote state equations and computed $\Phi(t)$ for each. For example, $1/s^2$ gives $\Phi(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$. I also noted which realisations share the same A matrix.

Problem 1-14 (controllability and observability). For several systems I built $E = [B\ AB\ \cdots]$ and $G = [C^T\ A^TC^T\ \cdots]$ and checked their ranks, which classifies each system as controllable or not and observable or not. I also drew the corresponding signal-flow graphs, including a companion-form system with $a_i \neq 0$ that is controllable with $\operatorname{rank} E = 4$.

Problem 1-15 (diagonal systems). For $\dot{x} = \operatorname{diag}(\lambda_1, \ldots, \lambda_4)x + bu$ and $y = c^Tx$ with distinct λ_i, the Vandermonde structure of E and G shows that the system is controllable iff every $b_i \neq 0$ and observable iff every $c_i \neq 0$.